

- 7 Part of a pattern of numbers is shown in the table below.

		C ₁	C ₂	C ₃	C ₄	C ₅	C ₆
Row	1	2	3	4	5	6	7
Row	2	8	9	10	11	12	13
Row	3	14	15	16	17	18	19
Row	4	20	21	22	23	24	25
Row	5	26	27	28	29	30	31
	*	*	*	*	*	*	*
	*	*	*	*	*	*	*
Row	n		x	y			
Row	$n+1$		s	t			

- (a) Express y , s and t in terms of x .

Answer

$y =$	
$s =$	
$t =$	

[2]

- (b) Show that the difference between xt and ys is always 6.

Answer

[2]

- (c) Express x in terms of n .

Answer $x =$ _____ [1]

- (d) Explain why xy would be in column number C_5 , for any value of n .

Answer _____ [2]

- (e) From the above, which column would ys lie in (C_1 , C_2 , C_3 , C_4 , C_5 or C_6)?

Answer _____ [1]

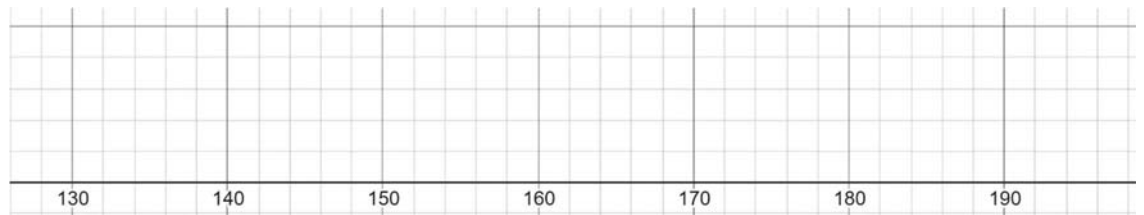
- 8 The ordered stem-and-leaf diagram records the height of 21 plants in Plantation A.

Stem	Leaf							
13	0	7	8	9	9			
14	1	1	2	6	7	8	9	9
15	0	2	3	5	6	8	8	8

Key: 14 | 8 means 148 cm

- (a) Represent the above information in a box and whisker.

Answer



[3]

- (b) State the modal height.

Answer _____ cm [1]

- (c) Calculate the percentage of plants having a height of less than 150 cm.

Answer _____ % [1]